RYAN ZERNACH

🎧 **AI Orchestration Engineering** 🤖 **Machine Learning Ops** 📱 **TypeScript React Native Apps**

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SUMMARY

I'm Ryan, an AI-savvy team member who ships production-grade software fast. I build agentic workflows supported by schema-validated outputs and lightweight evals. I measure impact in hours saved, p95 latency, accuracy, $/task, and $/quality. I've shipped both front-end apps & back-end infrastructure at every level of production environments. **10+ years of Engineering, 7+ years of Python, and 5+ years of TypeScript & React**

TECH STACK

- LLM Ops / Semantic Kernel / LangChain / CrewAI
- OpenAPI Pipelines / MCP / Agents / Evaluations
- Machine Learning / PyTorch / NLP / Fine-Tuning
- Cursor / Copilot / Windsurf / Continue
- Databases / SQL / GraphQL / NoSQL
- Azure / AWS / GCP / Cloud / SecOps
- GitHub Actions / CICD / FastLane
- DevOps / Docker / Kubernetes
- React Native / iOS / Android / Web / Expo Router
- TypeScript / React JS / Next JS / HTML / Streamlit
- Apple Pay / Payments / Subscriptions / Checkout
- IMAP / POP / Email Automations / Notifications
- Admin & Ops Dashboards / Internal CLI Tools
- WebSockets / Flask / FastAPI / Express / DNS
- Queues / RabbitMQ / Scheduled Cron Jobs
- Marketing Analytics / Sales Engineering

EXPERIENCE

AI Engineer & React Native Engineer • **Archlife Industries, LLC** *(100% Remote)* *Dec 2020 – Present*

- Published four social networking apps and two digital marketplace platforms
- Deployed agentic Python endpoints for facilitating merchant onboarding & inventory management
- Architected marketplace shopping platform with an emphasis on security, scalability, & analytics
- Shipped a cross between Robinhood & Facebook for securely exchanging stocks & crypto with scoreboards
- Automated email marketing campaigns to attract business owners to list their products on our platform
- Created a fitness iOS/Android app for managing gym sessions and making online payments to trainers
- Refined an iPad React Native app to improve the map's performance and the bluetooth extension

AI Engineer & React Native Engineer • **CellarTracker** *(100% Remote, Seattle)* Aug 2023 - July 2025

- Helped to boost the new app's ratings to 4.9 average, over 5,000+ 5-star ratings
- Wrote both the front-end & back-end for CellarChat™ LLM agent to 1M+ members
- Architected OpenAPI pipeline & LLM evaluation pipelines to judge metrics over time
- Embedded Apple Pay code in the React Native mobile app to generate higher revenue
- Rebuilt legacy mobile app using React Native to support both iOS & Android devices
- Created end-to-end (e2e) test suites to ensure long-term stability of key features
- Implemented activity feed posts, barcode scanning, image search, and analytics
- Iteratively improved the redux store for efficient and persistent state management
- Engineered the API for collecting mobile app report tickets, which generate HTML emails
- Developed the mobile app's analytics implementations for tracking tracking user interactions

React Native Engineer • **Demosphere** *(100% Remote, Washington D.C.)* Mar 2022 - May 2023

- Transitioned entire codebase from Javascript to Typescript for increased long-term maintainability
- Improved existing group chat features, push notifications, and native calendar sync to improve app reviews
- Evaluated Google Crashlytics to reduce edge case crashes and improve p95 stability & performance
- Improved computational and network efficiencies by reducing load times and runs to the database

React Native Engineer • **FinEquities** *(100% Remote, Miami)* *Feb 2021 – March 2022*

- Integrated various APIs, including: bank accounts, stocks, and cryptocurrencies
- Managed front-end, custom back-end cloud functions, & authentication endpoints
- Wrote the data visualizations logic for viewing the performance of stocks & crypto
- Built social network features: chat, user profile, posts, likes, emojis, and comments
- Constructed core functionality: copy portfolio, trending investors, and popular stocks
- Custom-built a companion admin web dashboard to monitor the mobile app's performance

College Tutor for Machine Learning, Web Development, & Data Science • **EdX** *(100% Remote)* *Oct 2020 – Dec 2021*

- Gained exposure to hundreds of machine learning applications through the act of teaching
- Taught 900+ college students machine learning, API's, JavaScript, HTML, Python, dashboards, & databases
- Managed a busy schedule of remotely teaching 30+ university students per week for ~60 minutes per lesson

Small Business Owner, Web Developer • **instaSod.com, CustomizedChristmasLights.com** *Oct 2015 – Jan 2019*

- Marketed my self-built websites and my businesses to generate $500k+ of gross sales in three years
- Planned & founded two LLC's in my early 20s: landscape design/contracting & professional christmas lighting
- Engineered user experience of uploading videos to website, answering questions, & receiving PDF pricing proposal

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EDUCATION

Data Science Software Engineering *Sept 2019 – Oct 2020*

Lambda School (AKA BloomTech) • 100% Remote • San Francisco, CA

- Productionizing Machine Learning Models, Python, Data Engineering, SQL, NoSQL, Back-End API's, Dashboards, Image Recognition Models, Natural-Language Processing (NLP) Models, Neural Networks, Data Visualizations, Statistical Analysis, Classic Algorithmic Paradigms, Time Complexity, Data Structures

Computer Science & Mathematics *Aug 2011 – Dec 2012*

Georgia Southern University • Statesboro, GA

- Calculus I, Calculus II, Calculus III, Statistics, Data Structures, Programming Principles I, Programming Principles II, Discrete Mathematics, Ethics of Computers in Society

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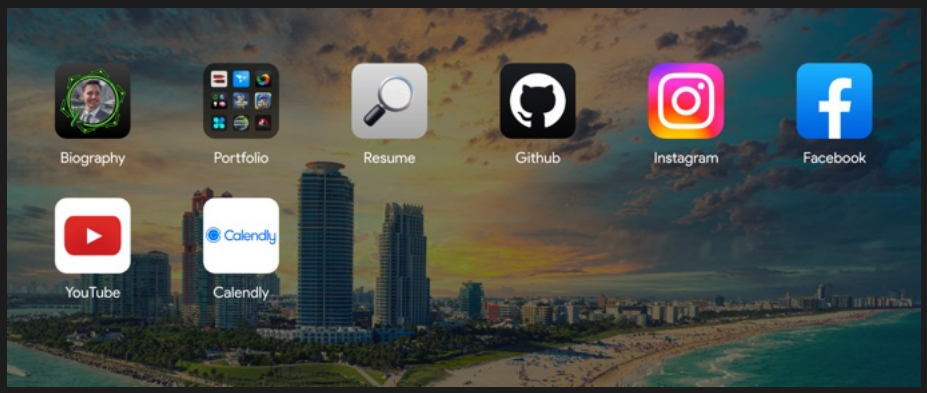
CERTIFICATIONS

- AWS Cloud Practitioner
- Google Cloud Developer
- DataRobot 10x Data Science AutoML
- Mosh: React Native Academy
- Y Combinator Startup School
- TensorFlow Developer

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